GLOBAL SCHOOL OF SCIENCE

Third Terminal Examination - 2082

Class: XI | **Subject: Mathematics (0071)** | **Full Marks: 75**

Time: 3 hrs. | **SET 'B'** | **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. **[11 × 1 = 11]**

1. The inverse of the statement $p \Rightarrow q$ is

   a) $\sim p \Rightarrow \sim q$ b) $\sim q \Rightarrow \sim p$

   c) $q \Rightarrow p$ d) $p \wedge \sim q$

2. A function $f: \mathbb{Z} - \{0\} \to \mathbb{Z}^+$ defined by $f(x) = |x|$ is

   a) one-one function b) onto function

   c) bijective function d) Neither

3. If $A = \begin{pmatrix} 2 & 2x+1 \\ x-1 & 3 \end{pmatrix}$ is a symmetric matrix, then $x =$

   a) 2 b) 3 c) -2 d) 0

4. The equation $x^2 + y^2 + 2px - 2qy - 1 = 0$ represents a pair of straight lines if

   a) $p^2 + q^2 = 1$ b) $p^2 + q^2 = -1$

   c) $p^2 - q^2 = 0$ d) $p^2 - q^2 = -1$

5. Let B_1 and B_2 be two bisectors of the angle between the lines AB and CD. If θ be the angle between B_2 and AB, then B_2 will be obtuse angle bisector if

   a) $|\tan\theta| > 1$ b) $|\tan\theta| < 1$

   c) $|\tan\theta| = \infty$ d) $\tan\theta = 1$

6. If the product of roots of the equation $mx^2 + 3x + 2m = 0$ is four times the sum of roots, then the value of m is

a) $\frac{-3}{2}$ b) $\frac{2}{3}$ c) 6 d) -6

7. The direction cosines of the y-axis are

a) $0, 0, 1$ b) $0, 1, 1$ c) $1, 0, 0$ d) $0, 1, 0$

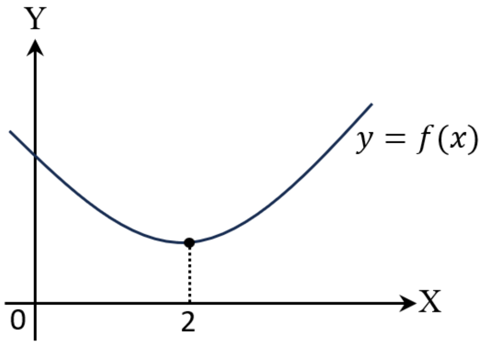
8. $\lim_{x \to \infty} \frac{\cos x}{x} =$

a) -1 b) 1 c) 0 d) does not exist.

9. If the function $f(x) = \begin{cases} \frac{\sin kx}{x} \\ 8 \end{cases}$ is continuous at $x = 0$, then $k =$

a) 8 b) -8 c) $\frac{1}{8}$ d) $\frac{-1}{8}$

10. Which of the following is true for the adjoining figure?



a) $f'(x) < 0$ for $(-\infty, 2)$, $f'(x) > 0$ for $(2, \infty)$ and $f''(x) < 0$ for all x.

b) $f'(x) < 0$ for $(-\infty, 2)$, $f'(x) > 0$ for $(2, \infty)$ and $f''(x) > 0$ for all x.

c) $f'(x) > 0$ for $(-\infty, 2)$, and $f''(x) > 0$ for all x.

d) $f'(x) < 0$ for $(2, \infty)$, and $f''(x) < 0$ for all x.

11. If $y = \ln(\log_a a^x)$ then $\frac{dy}{dx} =$

a) $\frac{1}{a^x . \ln a^x}$ b) $\frac{a^x}{\ln(a^x)}$ c) $\frac{1}{x}$ d) a^x

Group 'B'

Short Answer Questions. **[8 × 5 = 40]**

12. a) Construct the truth table for (i) $\sim (p \vee q)$ and (ii) $\sim p \wedge \sim q$. Are they logically equivalent? Explain. [1+1+1]

    b) For any $x, y \epsilon \mathbb{R}$, prove that: $|x + y| \leq |x| + |y|$. [2]

13. a) If $x = \log_a bc$, $y = \log_b ca$ and $z = \log_c ab$, then prove that: $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z} = 1$. [3]

    b) Let Q be the set of rational numbers and the function $f: Q \to Q$ is defined by $f(x) = 3x + 5$ for all $x \epsilon Q$. Does f^{-1} exist? [2]

14. a) Find inverse of the matrix $A = \begin{pmatrix} 3 & -2 & 1 \\ -1 & 1 & 0 \\ 2 & -2 & -1 \end{pmatrix}$ if exists. [3]

    b) Define absolute value of a complex number. If $z = \frac{(3,\ -4)}{(4,\ 3)}$ then find $|z|$. [1+1]

15. a) If p and q are the lengths of perpendiculars from origin upon the straight lines $x\, sec\, \theta + y\, cosec\, \theta = a$ and $x\, cos\, \theta - y\, sin\, \theta = a \cos 2\theta$ respectively, prove that: $4p^2 + q^2 = a^2$. [3]

    b) Find the value(s) of m so that a pair of equations $4x^2 + mx - 12 = 0$ and $4x^2 + 3mx - 4 = 0$ have a common root. [2]

16. a) Solve: $z^2 = -8 - 6i$ [3]

    b) Find the ratio in which the line joining the points $(-2, 4, 7)$ and $(3, -5, -8)$ is divided by the xy −plane. [2]

17. a) Evaluate: $\lim\limits_{x \to \theta} \frac{x \cot \theta - \theta \cot x}{x - \theta}$ [3]

    b) Define direction cosines of a line. Find the direction cosines of the line passing through the points $P(3, 2, 4)$ and $Q(5, 3, -1)$. [1+1]

18. a) Find, from first principle, the derivative of $f(x) = \sqrt{\cos 2x}$. [3]

    b) If $\cos(x + y) = \ln(x + y)$, then prove that: $\frac{dy}{dx} = -1$. [2]

19. a) Write the sufficient condition for a function $f(x)$ to have maximum value at $x = a$. For the function $f(x) = x + \frac{1}{x}$, show that maximum value is less than its minimum value. [1+2]

    b) Find $\frac{dy}{dx}$, when $y = \cos^{-1}(1 - 2x^2)$. [2]

Group ‘C’

Long Answer Questions. **[3 × 8 = 24]**

20. a) Let A, B, C be three non-empty sets. Prove that:
$A \cap (B - C) = (A \cap B) - (A \cap C)$ [2]

b) Find domain and range of the real valued function $f(x) = \frac{1}{\sqrt{x-5}}$. [2]

c) If $A = \begin{pmatrix} 2 & 3 \\ -4 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & -1 \\ 2 & -3 \end{pmatrix}$, then show that: $(AB)^T = B^T . A^T$. [2]

d) Prove that: $\begin{vmatrix} 1+a_1 & a_2 & a_3 \\ a_1 & 1+a_2 & a_3 \\ a_1 & a_2 & 1+a_3 \end{vmatrix} = 1 + a_1 + a_2 + a_3$

21. a) Prove that the straight lines joining the origin to the point of intersection of the line $y = mx + c$ and the circle $x^2 + y^2 = a^2$ are at right angles if $2c^2 = a^2(1 + m^2)$. [3]

b) Let two lines be $3x + 4y - 5 = 0$ and $4x + 3y + 7 = 0$.
i) Find the angle bisectors between them.
ii) Identify the bisectors containing origin. [1+1]

c) The sum of the roots of a quadratic equation is 1 and the sum of their squares is 13. Find the equation. [2]

d) Find the distance between two parallel lines $x \cos\theta + y \sin\theta - 2 = 0$ and $x \cos\theta + y \sin\theta - 3 = 0$. [1]

22. a) Let $y = f(x)$ be a function representing a curve. What does $\frac{dy}{dx}$ represent geometrically? [1]

b) For the function $f(x) = x^3 - 6x^2 + 9x + 15$, find
i) slope of tangent to the curve represented by above function at $(4, 1)$.
ii) global maximum value in $[-1, 3]$.
iii) point of inflection. [1+1+1]

c) Evaluate: $\lim_{x \to 25} \frac{x - 25}{\sqrt{x} - 5}$ [1]

d) A function $f(x)$ is defined as,
$$f(x) = \begin{cases} x^2 - 3 & for \quad x > 4 \\ 12 & for \quad x = 4 \\ 2x + 5 & for \quad x < 4 \end{cases}$$
Show that $f(x)$ has removable discontinuity at $x = 4$. Also, redefine $f(x)$ to make it continuous. [2+1]

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